National University
of Computer and Emerging Sciences
Chiniot-Faisalabad Campus

NS 1001 - Applied Physics

Q1:- Given:

$q = 70\,pC = 70 \times 10^{-12}\,C$
$V = 20V$

Solution:

a) $C = \frac{q}{V}$

$C = \frac{70 \times 10^{-12}}{20}$

$\boxed{C = 3.5 \times 10^{-12}\,F}$

b) We know that capacitance do not depend on charge So, the capacitance remain $C = 3.5 \times 10^{-12}\,F$

c) If $q = 200 \times 10^{-12}\,C$

then, $V = \frac{200 \times 10^{-12}\,C}{3.5 \times 10^{-12}}$

$\boxed{V = 57V}$

Q4: Given:

a radius $= 38 \times 10^{-3}\,m$
b radius $= 40 \times 10^{-3}\,m$

Sol:-

a) $C = 4\pi \varepsilon_0 \frac{ab}{b-a}$

$= 1.11 \times 10^{-10} \times \frac{(38 \times 10^{-3})(40 \times 10^{-3})}{(40 \times 10^{-3}) - (38 \times 10^{-3})}$

$\boxed{C = 8.436 \times 10^{-11}\,F}$

b) $C = \frac{A\varepsilon_0}{d} \Rightarrow A = \frac{Cd}{\varepsilon_0}$

$= \frac{8.436 \times 10^{-11} \times (40-38) \times 10^{-3}}{8.85 \times 10^{-12}}$

$\boxed{A = 191\ cm^2}$

Q7: Given:

$d_s = 10^{-12}\ m$

$V_s = 20\,V$

Solution:

$q = Ne$ — i)

$N = nAd$ put in i) $\Rightarrow q = nAde$ — ii)

$q = CV$ — iii)

comparing ii) and iii)

$CV = nAde \Rightarrow C/A = \frac{nde}{V}$

$= \frac{8.49 \times 10^{28} \times 10^{-12} \times 1.6 \times 10^{-}}{20}$

$\boxed{C/A = 6.79 \times 10^{-4}\ F/m^2}$.

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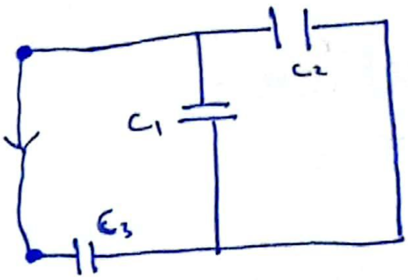
Q11: Given:

$C_1 = 10\,\mu F$
$C_2 = 5\,\mu F$
$C_3 = 4\,\mu F$

Solution:



$C_{12} = C_1 + C_2 = (10+5)\,\mu F = 15\,\mu F$

$\frac{1}{C_{eq}} = \frac{1}{C_{12}} + \frac{1}{C_3}$

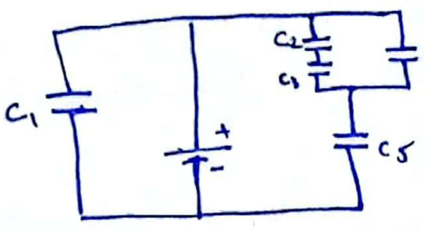
$= \frac{1}{15} + \frac{1}{4} = \frac{15+4}{60}$

$\frac{1}{C_{eq}} = \frac{19}{60} \Rightarrow C_{eq} = \frac{60}{19}$

$\boxed{C_{eq} = 3.1578\,\mu F}$

Q14: Given:

$V = 10V$
$C = 10\mu F$ for all capacitors



Solution:

a) $q_1 = C_1 V = 10 \times 10^{-6} \times 10 = 10^{-4} C$

$\boxed{q_1 = 10^{-4} C}$

b) $C_{23} = \frac{C_2 C_3}{C_2 + C_3} = \frac{100}{20} = 5\mu F$

$C_{234} = C_{23} + C_4 = 5\mu F + 10\mu F = 15\mu F$

$$C_{2345} = \frac{C_{234} C_5}{C_{234} + C_5} = 6\,\mu F$$

$$C_{eq} = C_1 + C_{2345} = 10\,\mu F + 6\,\mu F = 16\,\mu F$$

$$q_{2345} = C_{2345} V = 6\,\mu F \cdot 10V = 060\,\mu C$$

$$q_2 = C_{23} V_{23} = 5 \times \frac{60}{15} = 5 \times 4$$

$$V_{23} = \frac{Q_{234}}{C_{234}} = \frac{60}{15} = 4$$

$$\boxed{q_2 = 20\,\mu C}$$

Q27: Given:

$C_1 = 1\,\mu F$, $C_2 = 2\,\mu F$, $C_3 = 3\,\mu F$, $C_4 = 4\,\mu F$

Solution:

a) $q_1 = C_{13} V = \frac{C_1 C_3}{C_1 + C_3} V = \frac{1 \cdot 3}{1+3} \times 12 = \frac{3}{4} \times 12^3 = 9\,\mu C$

b) $q_2 = C_{24} V = \frac{C_2 C_4}{C_2 + C_4} \times 12 = \frac{8}{6} \times 12 = 16\,\mu C$

c) $q_3 = q_1 = 9\,\mu C$

d) $q_4 = q_2 = 16\,\mu C$

e) $V_1 = \frac{C_3 + C_4}{C_1 + C_2 + C_3 + C_4} V = \frac{3+4}{1+2+3+4} \times 12 = \frac{7}{10} \times 12 = 8.4\,V$

$q_1 = C_1 V_1 = 1 \times 8.4 = 8.4\,\mu C$

f) $q_2 = C_2 V_1 = 2 \times 8.4 = 16.8\,\mu C$

g) $q_3 = C_3 (12 - V_1) = 3 \times (12 - 8.4) = 10.8\,\mu C$

h) $q_4 = C_4 (12 - V_1) = 4 \times (12 - 8.4) = 14.4\,\mu C$